

Math 871 Topology I

Problem Set 4

1. Find the closure of $(0, 1)$ in $\mathbb{R}_\ell$.

Solution. Recall that the open sets of the form $[a, b)$ with $a < b$ form a basis for this topology. Consider any basic open neighborhood $U = [a, b)$ of 0, and note that we necessarily have $0 < b$. Thus $U \cap (0, 1) \neq \emptyset$, so $0 \in \overline{(0, 1)}$. Now for any real number $x < 0$, the open set $U = [x, 0)$ is a neighborhood of x but $U \cap (0, 1) = \emptyset$, so $x \notin \overline{(0, 1)}$. Finally, if $x > 1$, then $U = [x, x+1)$ is an open neighborhood of x but $U \cap (0, 1) = \emptyset$. We conclude that $\overline{(0, 1)} = [0, 1)$.

2. Let X be a topological space and consider $A \subseteq B \subseteq X$ with the subspace topologies.

- (a) Show that if A is dense in B and B is dense in X , then A is dense in X .

Solution. First, note that a subset S of Z is dense in Z if and only if every nonempty open set U in Z has $U \cap S \neq \emptyset$.

Suppose A is dense in B and B is dense in X , and consider any open set U in X . Since B is dense in X , $V := U \cap B \neq \emptyset$. Moreover, $U \cap B$ is by definition an open set in B (with the subspace topology). Note moreover that $A \cap B = A$ since $A \subseteq B$. Since A is dense in B and $U \cap B$ is a nonempty open set in B ,

$$U \cap A = U \cap (B \cap A) = (U \cap B) \cap A = V \cap A \neq \emptyset.$$

We have thus shown that every nonempty open set U in X intersects A nontrivially, and thus A is dense in X .

- (b) Show that if A is not dense in B , then A is not dense in X .

Solution. If A is not dense in B , then there some nonempty open set V of B in the subspace topology of B such that $V \cap A = \emptyset$. By definition of the subspace topology, $V = U \cap B$ for some open set U in X . Since V is nonempty, then so is U . Since $A \subseteq B$, then $A \cap B = A$, and

$$U \cap A = U \cap (B \cap A) = (U \cap B) \cap A = V \cap A = \emptyset.$$

Now U is a nonempty open set in X that does not intersect A . We conclude that A is not dense in X .

3. Let X and Y be topological spaces and consider $A \subseteq X$ and $B \subseteq Y$. Show that in $X \times Y$ with the product topology,

$$\overline{A \times B} = \overline{A} \times \overline{B}.$$

Solution. ($\subseteq$) Let $(x, y) \in \overline{A \times B}$, let U be a neighborhood of x in X , and let V be a neighborhood of y in Y . Then by the definition of the product topology on $X \times Y$, the product $U \times V$ is an open neighborhood of (x, y) in $X \times Y$. Since $(x, y) \in \overline{A \times B}$, any open neighborhood of (x, y) must intersect $A \times B$, and thus

$$U \times V \cap A \times B \neq \emptyset,$$

so there is some $(a, b) \in U \times V \cap A \times B$. Thus $a \in U \cap A$ and $b \in V \cap B$. Since U and V were arbitrary open neighborhoods of x and y respectively, we have shown that any open neighborhood U of x has $U \cap A \neq \emptyset$ and any open neighborhood V of y has $V \cap B \neq \emptyset$. Therefore, $a \in \overline{A}$ and $b \in \overline{B}$, and thus $(a, b) \in \overline{A} \times \overline{B}$. We conclude that

$$\overline{A \times B} \subseteq \overline{A} \times \overline{B}.$$

($\supseteq$) Let $(x, y) \in \overline{A} \times \overline{B}$, so that $x \in \overline{A}$ and $y \in \overline{B}$. Let $U \times V$ be a basic open neighborhood of (x, y) in $X \times Y$, so that U is an open neighborhood of x in X and V is an open neighborhood of y in Y . Since $x \in \overline{A}$, we must have $U \cap A \neq \emptyset$, so in particular there is some $a \in U \cap A$. Likewise, $V \cap B \neq \emptyset$, and there exists some $b \in V \cap B$. Therefore, $(a, b) \in (U \times V) \cap (A \times B)$. We showed that any basic open neighborhood $W = U \times V$ of (x, y) intersects $A \times B$. On the other hand, any general open neighborhood W of (x, y) is a union of basic open sets, at least one which is a neighborhood of (x, y) , and which must then intersect $A \times B$ nontrivially. We conclude that any open neighborhood W of (x, y) intersects $A \times B$ nontrivially, and thus $(x, y) \in \overline{A \times B}$. This shows that

$$\overline{A \times B} = \overline{A} \times \overline{B},$$

as desired.

4. Let X be a topological space and $A \subseteq X$. The **boundary** of A is defined by

$$\text{Bd}(A) = \overline{A} \cap \overline{X \setminus A}.$$

- (a) Show that A^{int} and $\text{Bd}(A)$ are disjoint.

Solution. Since any subset is contained in its closure, $X \setminus A \subseteq \overline{X \setminus A}$, and thus

$$X \setminus \overline{X \setminus A} \subseteq X \setminus (X \setminus A) = A.$$

Therefore, $X \setminus \overline{X \setminus A}$ is an open set (since it is the complement of a closed set) contained in A . Since A^{int} contains any open set contained in A , we conclude that

$$X \setminus \overline{X \setminus A} \subseteq A^{\text{int}}.$$

Thus $A^{\text{int}} \cap \overline{X \setminus A} = \emptyset$. Therefore,

$$A^{\text{int}} \cap \text{Bd}(A) = A^{\text{int}} \cap (\overline{A} \cap \overline{X \setminus A}) \subseteq A^{\text{int}} \cap \overline{X \setminus A} = \emptyset$$

as desired.

(b) Show that $\overline{A} = A^{\text{int}} \cup \text{Bd}(A)$.

Solution. ($\subseteq$) Let $x \in \overline{A}$. If $x \notin \text{Bd}(A)$, then $x \notin \overline{X \setminus A}$. We showed in (a) that

$$X \setminus \overline{X \setminus A} \subseteq A^{\text{int}}.$$

Thus $x \in A^{\text{int}}$. We conclude that $\overline{A} \subseteq A^{\text{int}} \cup \text{Bd}(A)$.

($\supseteq$) We have

$$A^{\text{int}} \subseteq A \subseteq \overline{A}$$

and

$$\text{Bd}(A) = \overline{A} \cap \overline{X \setminus A} \subseteq \overline{A}.$$

Then

$$A^{\text{int}} \cup \text{Bd}(A) \subseteq \overline{A}.$$

(c) Show that $\text{Bd}(A) = \emptyset$ if and only if A is a clopen set.

Solution. ($\implies$) If $\text{Bd}(A) = \emptyset$, then by (b)

$$\overline{A} = A^{\text{int}}.$$

In general,

$$A^{\text{int}} \subseteq A \subseteq \overline{A},$$

so we conclude these must all be equalities. In particular, $A = \overline{A}$ is closed, and $A = A^{\text{int}}$ is open.

($\impliedby$) By contrapositive. Suppose that $\text{Bd}(A) \neq \emptyset$. By (a), A^{int} and $\text{Bd}(A)$ are disjoint, so it follows that

$$\overline{A} = A^{\text{int}} \cup \text{Bd}(A) \supsetneq A^{\text{int}}.$$

Since $A^{\text{int}} \subseteq A \subseteq \overline{A}$ and $A^{\text{int}} \subsetneq \overline{A}$, we have that $A^{\text{int}} \subsetneq A$ or $A \subsetneq \overline{A}$, or possibly both. If $A^{\text{int}} \subsetneq A$, then A is not open. If $A \subsetneq \overline{A}$, then A is not closed. In particular, A is not clopen.